

BAREC-TB: A Dependency Treebank and Parsing Benchmark for Arabic Children’s Texts

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Abstract

We present BAREC-TB, the first dependency treebank of Arabic children’s literature and educational texts. The 156K-word corpus includes texts drawn from eight sources, including curricula, reading materials, children’s stories, poems, and songs. The treebank is annotated according to the Columbia Arabic Treebank (CATiB) dependency representation. We compare BAREC-TB with existing Arabic treebanks, including the Penn Arabic Treebank and Camel Treebank, using corpus statistics, dependency-based measures of syntactic complexity, readability and vocabulary-related features. We also establish parsing benchmarks using both a trainable parser and large language models. The corpus will be made publicly available to support Arabic NLP, educational NLP, and language pedagogy research.¹

1 Introduction

Arabic children’s literature and educational texts are an important yet underrepresented domain in Arabic NLP. These texts play a central role in classroom instruction, reading practice, and language learning. They also differ from adult-oriented Arabic in ways that are directly relevant to NLP: they often contain controlled vocabulary, shorter sentences, dialogues, questions, poems, songs, and pedagogically motivated discourse structures (Taha-Thomure, 2017; Habash et al., 2025). Syntactically annotated resources for this domain can support educational NLP, readability assessment, language pedagogy, and research on literacy development by providing evidence about the linguistic constructions presented to young readers at different stages.

This gap is especially visible in Arabic treebanking and dependency parsing. While Arabic has several major treebanking efforts, existing resources largely target adult-oriented text and do not provide

dedicated coverage of children’s literature or educational materials (MacWhinney, 2000; Maamouri et al., 2004; Habash and Roth, 2009; Dukes and Buckwalter, 2010; Al-Ghamdi et al., 2021; Habash et al., 2022; Elshabrawy et al., 2023; Khalifa et al., 2026). As a result, parsers trained on existing Arabic treebanks may not generalize well to children’s materials because of domain differences in genre, sentence style, and syntactic distribution. This also makes child-directed Arabic a useful test case for large language models (LLMs). Prior work shows that their structured linguistic analyses are sensitive to prompting, tokenization, and morphosyntactic complexity (Adel et al., 2026). A dedicated dependency treebank of Arabic children’s text therefore provides both a needed resource for studying child-directed Arabic syntax and a benchmark for evaluating trainable parsers and LLM-based parsing.

In this paper, we present **BAREC-TB**, the first dependency treebank dedicated to Arabic children’s literature and educational texts. BAREC-TB contains approximately 156K words drawn from eight source types, including curricula, reading materials, children’s stories, poems, and songs, all selected from the Foundational subset of The Balanced Arabic Readability Evaluation Corpus (BAREC) (Elmadani et al., 2025b). The treebank follows the Columbia Arabic Treebank (CATiB) dependency representation (Habash and Roth, 2009), extending an established Arabic treebanking framework to a new child-directed and educational domain. An example of a syntactic tree is shown in Figure 1.

Our contributions are: (a) We **introduce BAREC-TB**, the first dependency treebank dedicated to Arabic children’s texts. (b) We **compare BAREC-TB with existing Arabic treebanks** using corpus statistics, syntactic complexity measures, readability and vocabulary-related features. (c) We **establish parsing benchmarks** using both trainable parsers and LLMs.

¹<https://barec.camel-lab.com/>

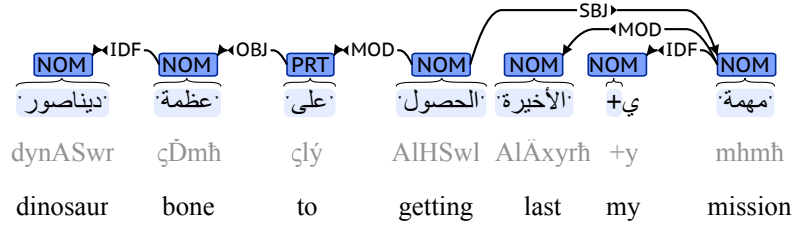


Figure 1: Example CATiB dependency tree for “My last mission is to get a dinosaur bone” (Majed Magazine, 2011). Words are shown with transliteration and gloss.

2 Background and Related Work

2.1 Arabic Treebanks

Arabic has several syntactically annotated resources that vary in size, domain, and annotation formalism. The Penn Arabic Treebank (PATB) is one of the most widely used Arabic treebanks (Maamouri et al., 2004). It was constructed primarily from news and web sources, and annotated using phrase structure. It has since been converted to dependency representations, including CATiB (Habash and Roth, 2009) and Universal Dependencies (UD) (Taji et al., 2017). The Prague Arabic Dependency Treebank (PADT) was the earliest Arabic dependency treebank (Srnč et al., 2002) and was later extended to UD. The Columbia Arabic Treebank (CATiB) was annotated directly in a dependency formalism inspired by traditional Arabic grammar (Habash et al., 2009), and Camel Treebank extends CATiB-style annotation to a multi-genre collection of Classical Arabic and Modern Standard Arabic texts (Habash et al., 2022). Other resources target specialized domains, such as the Quranic Arabic Corpus (Dukes and Buckwalter, 2010; Dukes et al., 2010), the *i3rab* treebank (Halabi et al., 2021), and the Arabic Poetry Treebank (ArPoT) (Al-Ghamdi et al., 2021). BAREC-10M includes children’s texts among its 10 million words of synthetically generated syntactic trees (Elmadani et al., 2026). BAREC-TB addresses this limitation with manually annotated dependency trees for Arabic children’s texts.

Despite this progress, existing Arabic treebanks mostly cover adult-directed material, including news, web text, religious scripture, poetry, and general-domain data. Children’s texts remain largely absent from manually annotated Arabic dependency resources. BAREC-TB addresses this gap by providing the first dependency treebank dedicated to Arabic children’s texts.

2.2 Children’s Language Resources

Children’s language resources support educational NLP, readability assessment, text simplification, and language pedagogy. Large multilingual resources such as CHILDES (MacWhinney, 2000) provide extensive child-directed speech corpora, with substantial work on syntactic parsing for English child language (Sagae et al., 2007; SAGAE et al., 2010; Liu and Prud’hommeaux, 2023; Szubert et al., 2025; Yang et al., 2025). In English, graded reading systems such as Fountas and Pinnell’s 27-level scale are widely used to match texts to children’s reading abilities (Fountas and Pinnell, 2006). For Arabic, Taha-Thomure (2017) proposed a 19-level leveling framework for children’s books, using criteria such as genre, vocabulary, sentence structure, diacritization, illustrations, and word count.

Several Arabic resources target readability and educational materials. Early Arabic readability models relied on surface features such as word and sentence length (Al-Dawsari, 2004; Al-Khalifa and Al-Ajlan, 2010), while later work incorporated POS, morphology, orthographic features, and pretrained models (Forsyth, 2014; Saddiki et al., 2018; El-Haj and Rayson, 2016; Liberato et al., 2024). Resource-building efforts include SAMER (Al Khalil et al., 2020; Alhafni et al., 2024), CEFR-oriented Arabic datasets and vocabulary resources (Habash and Palfreyman, 2022; Naous et al., 2024; Soliman and Familiar, 2024), DARES (El-Haj et al., 2024; Almujaivel et al., 2025), and BAREC, a fine-grained sentence-level readability corpus across 19 levels (Elmadani et al., 2025b).

2.3 Dependency Parsing for Arabic

Arabic dependency parsing has progressed from feature-rich and transition-based systems to neural models that use pretrained language representations (Green and Manning, 2010; Marton

et al., 2013; Shahrour et al., 2016; Straka, 2018; Kankanampati et al., 2020; Al-Ghamdi et al., 2023). A recent example is CAMELParser2.0 (Elshabrawy et al., 2023), a strong BERT-based supervised parser for Arabic dependency parsing. Its compatibility with the CATiB representation makes it a suitable baseline for evaluating parsing performance on BAREC-TB.

Recent work has begun evaluating large language models (LLMs) on structured Arabic linguistic analysis. In particular, Adel et al. (2026) evaluate LLMs on Arabic morphosyntactic tagging and dependency parsing. We build on this line of work by evaluating LLMs under their strongest reported settings, while also comparing them with trainable supervised parsers.

2.4 BAREC Project

BAREC-TB is part of the broader BAREC project, which develops resources for Arabic readability across the Arab world. The project began with the 19-level BAREC Annotation Framework (Habash et al., 2025), followed by the BAREC Corpus (Elmadani et al., 2025b), a large-scale manually annotated sentence-level readability dataset. BAREC Demo (Altarouch et al., 2025) made this work accessible through a web-based Arabic readability assessment system, while the BAREC Shared Task 2025 (Elmadani et al., 2025a) supported community benchmarking. BAREC-10M (Elmadani et al., 2026) further expanded the project with a large automatically labeled corpus. BAREC-TB extends this line of work by adding dependency parsing annotations for the child-directed portion of BAREC.

3 BAREC-TB Corpus

To ensure that the treebank focuses on children-oriented texts, we select documents from the Foundational level of the BAREC corpus (Elmadani et al., 2025b). This subset contains materials intended for early and developing readers (up to 4th grade or age 10), making it well suited for constructing a dependency treebank of Arabic children’s texts. In total, BAREC-TB covers 50% of the words in the Foundational portion of BAREC. Our choice of BAREC is further motivated by its sentence-level readability annotations, which make it possible to study the relationship between syntactic structure and readability. We preserve the original BAREC train, development, and test splits

Split	#Docs		#Sents		#Words	
Train	193	60%	9,580	62%	96,054	62%
Dev	62	19%	3,071	20%	29,510	19%
Test	65	20%	2,732	18%	30,466	20%
BAREC-TB	320	100%	15,383	100%	156,030	100%

Table 1: Statistics for the BAREC-TB splits.

for all selected documents. Table 1 shows the corpus splits in terms of documents, sentences, and words.

3.1 Data Sources

We select the corpus texts from eight resources (presented in order of sub-corpus word size):

Curriculum A portion of the UAE curriculum textbooks for the grades from 1 to 4 in three subjects: Arabic language, social studies, and Islamic studies (Khalil et al., 2018).

Majed A subset of 10 manually typed editions of Majed magazine for children from 1983 to 2019.²

Green Library 30 manually typed books from the Green Library.³

Hindawi A subset of 30 children stories books extracted from the Hindawi Foundation website.⁴

ArabicMMLU 878 question and answer pairs from the primary level of the ArabicMMLU benchmark dataset (Koto et al., 2024).

Collection of Children’s Poems (Other) Example of the included poems: My Language Sings (لغتي تغني), and Poetry and News (أشعار وأخبار) (Al-Safadi, 2005; Taha-Thomure, 2007).

Spacetoons The opening songs of 53 animated children series from Spacetoons channel.

ChatGPT 443 sentences generated by ChatGPT and limited to children’s vocabulary. All sentences were manually validated for correctness.⁵

3.2 Corpus Statistics

Table 2 presents corpus statistics for BAREC-TB and its source categories, together with PATB (Maamouri et al., 2004) and CamelTB (Habash et al., 2022).

²https://archive.org/details/majid_magazine

³https://archive.org/details/201409_201409

⁴<https://www.hindawi.org/books/categories/>

⁵<https://chatgpt.com/>

Source	Docs	Sents	Words	Tokens	Found.	Adv.	Spec.	Genre	W/S	T/S
Curriculum	38	5,361	39,113	46,166	100%	0%	0%	Education	7.3	8.6
Majed	98	3,954	38,386	46,152	100%	0%	0%	Comics	9.7	11.7
Green Library	30	1,410	23,248	29,412	100%	0%	0%	Stories	16.5	20.9
Hindawi	30	1,582	22,716	28,116	100%	0%	0%	Stories	14.4	17.8
ArabicMMLU	49	878	22,334	23,558	100%	0%	0%	Education	25.4	26.8
Other	19	886	3,890	4,664	100%	0%	0%	Poetry	4.4	5.3
Spacetoan	53	869	3,841	4,606	100%	0%	0%	Poetry	4.4	5.3
ChatGPT	3	443	2,502	2,878	100%	0%	0%	Education	5.6	6.5
BAREC-TB	320	15,383	156,030	185,552	100%	0%	0%	Children	10.1	12.1
PATB	1,834	19,738	628,656	738,889	0%	100%	0%	News	31.9	37.4
CamelTB	535	13,337	187,736	241,910	0%	56%	44%	Multi	14.1	18.1

Table 2: Corpus statistics for BAREC-TB sources, CamelTB and PATB. Found., Adv., and Spec. indicate the percentage of words in the Foundational, Advanced, and Specialized levels, respectively. W/S and T/S indicate average words and tokens per sentence, respectively.

The three treebanks differ clearly in readership level distribution: Foundational targets early readers building basic literacy, Advanced targets average adult reading, and Specialized targets complex domain-specific reading. All BAREC-TB texts belong to the Foundational level, reflecting its focus on children’s text. In contrast, PATB is entirely Advanced, while Camel Treebank contains both Advanced and Specialized texts. This shows that BAREC-TB fills the child-directed portion of the Arabic treebanking landscape that is not covered by the other resources.

Table 2 also shows clear differences in sentence length. BAREC-TB has the lowest average sentence length, with 10.1 words and 12.1 tokens per sentence, compared with 31.9 words and 37.4 tokens in PATB and 14.1 words and 18.1 tokens in Camel Treebank. This is expected, since children’s texts tend to use shorter sentences than adult-oriented text.

Within BAREC-TB, ArabicMMLU has the highest average sentence length, with 25.4 words and 26.8 tokens per sentence, because each sentence represents a question together with its answer options; excluding ArabicMMLU, the BAREC-TB average sentence length drops to 9.2 words and 11.2 tokens per sentence. The story sources follow, with Green Library averaging 16.5 words and 20.9 tokens per sentence and Hindawi averaging 14.4 words and 17.8 tokens per sentence. Poetry and song sources have the shortest sentences, which explains the low averages for Other Minor Sources and Spacetoan Songs, both at 4.4 words and 5.3 tokens per sentence.

4 Annotation Framework

4.1 CATiB Dependency Representation

We annotate BAREC-TB using the Columbia Arabic Treebank (CATiB) dependency representation (Habash and Roth, 2009). CATiB is well suited for Arabic treebanking because its design is influenced by traditional Arabic grammar, which makes the representation easier for annotators familiar with Arabic linguistic terminology.

Tokenization CATiB adopts the tokenization conventions used in earlier Arabic treebanking efforts, PATB and PADT (Maamouri et al., 2004). Under this scheme, all clitics are separated into independent tokens, except for the definite article **ال** *Al*+ ‘the’.

POS Tags CATiB has six POS tags: **NOM** (non-proper nominals including nouns, pronouns, adjectives and adverbs), **PROP** (proper nouns), **VRB** (active-voice verbs), **VRB-PASS** (passive voice verbs), **PRT** (particles such as prepositions or conjunctions) and **PNX** (punctuation). Following CamelTB (Habash et al., 2022), we also use **FOREIGN** for tokens written in non-Arabic script.

Relations CATiB’s dependency links are labeled with one of eight relation labels: **SBJ** (subject of verb or topic of simple nominal sentence), **OBJ** (object of verb, preposition, or deverbal noun), **TPC** (topic in complex nominal sentences containing an explicit pronominal referent), **PRD** (predicate marking the complement of the extended copular constructions), **IDF** (relation between the possessor [dependent] to the possessed [head] in the

idafa/possessive nominal construction), **TMZ** (relation of the specifier [dependent] to the specified [head] in the tamyiz/specification nominal constructions), **MOD** (general modifier of verbs or nouns), and **—** (marking flatness inside constructions such as first-last proper name sequences).

4.2 Annotation Workflow

Our annotation workflow follows the pipeline used for Camel Treebank (Habash et al., 2022): sentence segmentation, then automatic parsing, followed by manual correction of tokenization, POS tags, and dependency relations. Since BAREC-TB is selected from BAREC corpus, which is already sentence-segmented, we do not perform an additional sentence segmentation step.

Automatic Annotation We use Camel-Parser2.0 (Elshabrawy et al., 2023) for the initial automatic parsing. Within the parser component, we use the morphological disambiguation component of *CAMEL Tools* (Obeid et al., 2020; Inoue et al., 2022)⁶ with SAMA database (Graff et al., 2009) for tokenization and POS tagging.

Manual Annotation The annotation team consists of two Arabic native speakers with extensive treebanking experience. Annotation is carried out using Palmyra3.0 (AbuOdeh et al., 2024),⁷ a configurable, platform-independent tool for visualizing and editing dependency trees. It allows annotators to correct tokenization, POS tags, and dependency relations in a single interface.

Annotators’ Observations Our annotation team observed that most annotation decisions in BAREC-TB were comparable to those encountered in newswire, literary, and other adult-oriented texts they previously worked on despite differences in genre and audience. In fact, syntactic annotation was sometimes simpler because children’s texts tend to employ shorter sentences, more canonical word order, greater repetition, and fewer structurally ambiguous constructions, especially in sources like the Green Library and Hindawi. The main challenges arose from educational and interactive formats that weaken the correspondence between surface presentation and syntactic structure. Elements such as fill-in-the-blank exercises, matching tasks, tables, lists, multiple-choice questions, and comic-style layouts often introduce omitted

Source	Docs	Sents	Tokens	UAS	LS	LAS
Curriculum	4	422	3,843	97.1	97.8	95.9
Majed	4	236	2,708	94.0	96.1	92.5
Green Library	5	269	5,195	95.4	97.3	94.2
Hindawi	4	153	2,720	97.2	98.9	96.5
ArabicMMLU	4	80	2,166	96.9	97.0	95.5
Other	1	39	557	98.4	97.3	96.6
Spacetoan	2	45	301	94.3	96.1	92.6
ChatGPT	1	152	827	99.9	99.8	99.6
Overall	25	1,396	18,317	96.3	97.5	95.1

Table 3: Per-source inter-annotator agreement (IAA). The overall scores are **token-weighted averages** across all sources.

constituents, unconventional sentence segmentation, and punctuation serving instructional or graphical rather than syntactic functions. This is reflected in the substantially higher frequency of punctuation in BAREC-TB and its frequent departure from the discourse- and syntax-oriented usage typical of conventional prose. Consequently, annotators could not consistently rely on punctuation, line breaks, or layout cues, and instead prioritized underlying grammatical relations when determining syntactic structure.

4.3 Inter-Annotator Agreement

To assess annotation consistency, we conducted two rounds of inter-annotator agreement (IAA). In the first round, two annotators independently annotated a subset of six BAREC-TB documents, then met to discuss disagreements and subsequently reviewed the annotations separately. Agreement improved from 95.6 UAS, 97.3 LS, and 94.3 LAS to 98.5 UAS, 99.2 LS, and 98.2 LAS,⁸ allowing us to refine the annotation guidelines before continuing with the remaining corpus. Then, we conducted a second and final IAA evaluation on a larger set of 19 documents. The annotators achieved 96.4 UAS, 97.4 LS, and 95.2 LAS, corresponding to error reduction rates of 17.2%, 5.9%, and 16.5% over the first-round agreement for UAS, LS, and LAS, respectively.

Table 3 reports per-source inter-annotator agreement on the full IAA set. Agreement is consistently high across all sources, with LAS ranging from 92.5 to 99.6. The lowest agreement is observed for Majed and Spacetoan, while Green Library and Hindawi achieve higher agreement. These results are consistent with the annotation challenges dis-

⁶*CAMEL Tools* v1.5.5: BERT-Disambig.

⁷<https://palmyra.camel-lab.com/>

⁸These dependency parsing metrics are explained in §6.1.

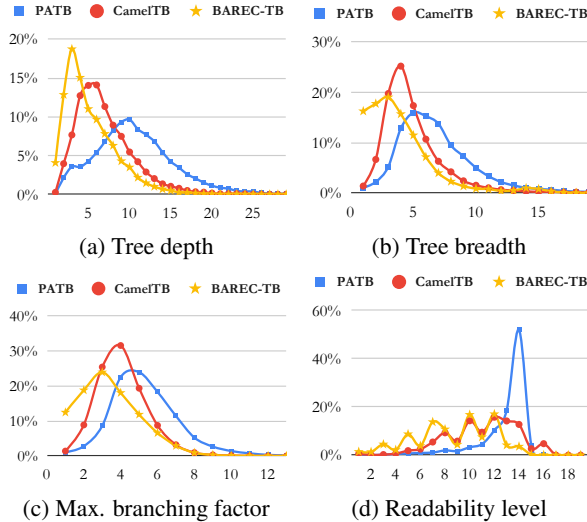


Figure 2: Distributions of tree depth, tree breadth, maximum branching factor, and readability level in PATB, CamelTB, and BAREC-TB.

cussed in § 4.2: sources with more interactive or artistic formats exhibit lower agreement than regular texts.

5 Analysis of BAREC-TB

5.1 Tree Structure and Readability

Figure 2 compares BAREC-TB sentences with PATB and CamelTB in terms of tree depth, tree breadth, maximum branching factor, and readability level. For BAREC-TB, we obtain the readability levels from BAREC Corpus, while for CamelTB and PATB we predict them using Elmadani et al.’s (2025b) best readability assessment system, the AraBERTv02+D3Tok+Reg model.⁹

Across the syntactic measures, BAREC-TB is concentrated at lower values, indicating shallower and less structurally dense trees. PATB shows the longest tail, especially for tree depth, while CamelTB generally falls between PATB and BAREC-TB. The readability distribution shows a similar contrast. BAREC-TB is concentrated in lower and middle readability levels, reflecting its focus on children’s texts. PATB is concentrated at higher levels, consistent with its adult-oriented news domain, while CamelTB spans a broader range. Overall, these patterns show that BAREC-TB differs from existing Arabic treebanks both syntactically and in readability profile.

⁹<https://huggingface.co/CAMEL-Lab/readability-arabertv2-d3tok-reg>

BAREC-TB Source	CamelTB	PATB
Hindawi	98.7% (1)	98.7% (1)
Majed	97.4% (3)	98.1% (2)
Green Library	97.8% (2)	97.1% (4)
Curriculum	96.4% (4)	97.8% (3)
ChatGPT	93.8% (6)	92.9% (5)
Spacetoan	95.0% (5)	92.2% (6)
Other	92.3% (7)	91.4% (7)
ArabicMMLU	77.7% (8)	83.7% (8)
All	96.4%	97.6%

Table 4: Correlation of POS and dependency-relation distributions between each BAREC-TB source and the corresponding distributions in CamelTB and PATB. Numbers in parentheses indicate the source rank by correlation.

Subtree Structure Overlap We further compare the treebanks using local subtree structures. We define a subtree structure as a parent node with its immediate children, matching only when the parent POS, ordered child POS tags, and dependency relations are identical. Across PATB, CamelTB, and BAREC-TB, we find 21,084 unique subtree structures, but only 1,770 (8%) appear in all three treebanks. Although this shared set is small in terms of unique types, it accounts for most observed subtree tokens in PATB and CamelTB (93%) and slightly less in BAREC-TB (90%), suggesting a common core of frequent Arabic syntactic patterns alongside more domain-specific structures in children’s texts.

Variation Across BAREC-TB Sources Table 4 groups BAREC-TB sources by their correlation with CamelTB and PATB **POS-relation distributions**. The highest-correlating sources are naturally structured children’s texts, including stories, comics, and curricula, suggesting that their basic syntactic profiles remain close to existing Arabic treebanks. The middle group reflects more source-specific structures: ChatGPT contains short educational sentences, while Spacetoan and Other minor sources mainly contain songs and poems. ArabicMMLU is the most distinct source, since its sentences consist of questions and answer options rather than continuous text. Overall, the table shows that BAREC-TB combines sources that are broadly similar to the existing Arabic treebanks with others that introduce genre-specific structures characteristic of children’s texts.

SAMER Level	BAREC-TB	CamelTB	PATB
1	24%	16%	12%
2	16%	12%	10%
3	21%	18%	15%
4	20%	22%	22%
5	10%	18%	22%
OOV	8%	14%	19%

Table 5: Distribution of lemma types across SAMER readability levels and OOV rate.

5.2 Lexical Analysis

We compare the lemma-level vocabularies of BAREC-TB, PATB, and CamelTB. All words are lemmatized in context using the best lemmatization system of [Saeed and Habash \(2025\)](#) with the SAMA analyzer ([Graff et al., 2009](#)), producing a shared inventory of SAMA and Out Of Vocabulary (OOV) lemmas for comparing overlap, coverage, and readability.

BAREC-TB contains 9,973 lemma types: 6,736 (68%) also occur in CamelTB, 7,214 (72%) in PATB, and 5,790 (58%) in all three treebanks. However, 1,813 BAREC-TB lemmas (18%) occur in neither CamelTB nor PATB, indicating both a substantial shared core and vocabulary specific to children’s texts.

To control for corpus size, we randomly sample CamelTB and PATB to match BAREC-TB. OOV rates are low across all three treebanks: 1.0% for BAREC-TB, 1.5% for CamelTB, and 0.9% for PATB. CamelTB and PATB nevertheless cover 33% and 21% more lemma types than BAREC-TB, respectively. Lexical diversity, measured as lemma types divided by total lemmas, is lowest in BAREC-TB (6.4%), compared with CamelTB (8.5%) and PATB (7.7%), reflecting the repetition expected in children’s texts.

Finally, we examine lemma readability using the SAMER readability lexicon ([Al Khalil et al., 2020](#)), which assigns SAMA lemmas to one of five readability levels. As shown in Table 5, BAREC-TB has a larger share of easier lemmas, with 40% of its lemma types in levels 1–2, compared with 28% in CamelTB and 22% in PATB. It also has fewer difficult and OOV lemmas, with 10% at level 5 and 8% OOV, compared with 18% and 14% in CamelTB, and 22% and 19% in PATB. These patterns confirm that BAREC-TB is lexically aligned with child-directed materials.

6 Parsing Benchmark

6.1 Experimental Setup

We evaluate Arabic dependency parsing on BAREC-TB using the CATiB dependency representation ([Habash and Roth, 2009](#)). In CATiB-style parsing, each token is assigned a syntactic head and a dependency relation label. Following prior works on Arabic dependency parsing, we evaluate systems under both gold-tokenized and raw-text input conditions.

Data We use BAREC-TB as the target treebank and evaluate all systems on its Dev and Test splits. Our experiments also involve using CamelTB and PATB for training. We consider three data configurations: BAREC-TB only, PATB+CamelTB, and PATB+CamelTB+BAREC-TB. These settings allow us to compare in-domain training, cross-domain transfer from existing Arabic treebanks, and mixed-domain training.

Supervised Parser For supervised dependency parsing, we follow CamelParser2.1 ([Adel et al., 2026](#)), which extends CamelParser2.0 ([Elshabrawy et al., 2023](#)) with projective multiroot decoding. Whereas CamelParser2.0 enforces exactly one root per input, CamelParser2.1 permits multiple roots while requiring at least one root and prohibiting cycles and self-loops.

The parser follows a pipeline design in which raw input is first preprocessed and morphologically disambiguated using CAMEL Tools ([Obeid et al., 2020](#)), while already tokenized input passes directly to the parser. The dependency parsing component is implemented using SuPar ([Zhang et al., 2020](#)), following the biaffine dependency parsing framework of [Dozat and Manning \(2017\)](#), and uses contextualized BERT representations instead of static word embeddings. Following the CamelParser2.0 setup, token-level representations are derived from WordPiece embeddings, and the parser predicts dependency arcs and labels over the tokenized sentence. In our experiments, we use CamelBERT-MSA ([Inoue et al., 2021](#)) as the pretrained encoder for all supervised parser configurations. We train CamelParser2.1 under the three data configurations described above: BAREC-TB, PATB+CamelTB, and PATB+CamelTB+BAREC-TB.

Large Language Models We also evaluate a set of strong LLMs spanning different model families

and access regimes. Our evaluation includes closed-source multilingual models, open-weight multilingual models, and Arabic-centric models: Gemini 3.5 Flash (Team et al., 2025), GPT-5.4 (Singh et al., 2026), Llama-4-Scout (Meta, 2025), and Fanar-2 (TEAM et al., 2026), henceforth **Gemini**, **GPT**, **Llama**, and **Fanar**, respectively. Each model is evaluated in a zero-shot setting and in a retrieval-based 10-shot in-context learning setting.

LLM Prompting and Retrieval For LLM-based parsing, we adopt the dependency parsing prompt formulation of Adel et al. (2026). The prompt explicitly defines the CATiB dependency labels, gives concise attachment and sentence-structure guidelines, and instructs the model to output only the required structured parse without explanations or extra fields. To ensure consistent machine-readable outputs, we use JSON-structured output interfaces for all LLM backends: the structured output modules of the OpenAI and Gemini APIs for proprietary models, and vLLM’s structured decoding support for open-source models (Kwon et al., 2023). In the gold-tokenized setting, the model is instructed to preserve the provided token sequence. In the raw-text setting, the model is instructed to first tokenize the input according to CATiB conventions and then produce the dependency tree.

For the 10-shot setting, we follow the retrieval-based in-context learning approach of Adel et al. (2026). For each evaluation sentence, we retrieve examples from the corresponding training set using only the input side of the training examples, ensuring that no development or test labels are used during retrieval. We select the ten examples with the highest chrF++ similarity to the input sentence. We evaluate three retrieval pools: BAREC-TB, PATB+CamelTB, and PATB+CamelTB+BAREC-TB. Since chrF++ is based on character n-gram overlap, this retrieval strategy favors examples that are close in surface form and can capture shared orthographic, lexical, and morphological patterns, which are especially relevant for Arabic. Model licenses and details are in Appendices A and B.

Evaluation Metrics We report Labeled Attachment Score (LAS), the percentage of tokens with the correct head and dependency label, as our primary metric. We also report Unlabeled Attachment Score (UAS), the percentage of tokens assigned the correct head, and LS, the percentage of tokens assigned the correct dependency label. For raw-text experiments, we report tokenization F1 (Tok).

6.2 Results

Table 6 presents the parsing results on the development and test sets. Overall, CamelParser2.1 remains the strongest system. Its best configuration, trained on PATB+CamelTB+BAREC-TB, achieves 95.5 LAS on the test set with gold tokenization and 95.0 LAS from raw text. This setting consistently outperforms both BAREC-TB-only training and PATB+CamelTB training, showing that existing Arabic treebanks provide strong syntactic supervision while BAREC-TB adds useful in-domain adaptation.

Among LLMs, Gemini is the strongest model across both splits and input conditions. With 10 examples retrieved from BAREC-TB, it reaches 92.9 LAS on the gold-tokenized test set and 92.2 LAS in the raw-text setting, narrowing the gap with the best supervised parser to 2.6 and 2.8 LAS, respectively. GPT is the second-best LLM, while Llama and Fanar remain substantially weaker.

The comparison between zero-shot and 10-shot settings shows that retrieved demonstrations are crucial for LLM parsing. On the test set with gold tokenization, Gemini improves from 76.0 to 92.9 LAS, GPT from 57.6 to 88.9, Llama from 28.1 to 77.2, and Fanar from 15.9 to 72.0. These large gains indicate that LLM performance is highly conditioned by the availability of similar annotated examples, and should therefore be interpreted as retrieval-augmented in-context parsing.

The retrieval source also matters. For all LLMs, BAREC-TB-only retrieval gives the best or tied-best 10-shot results across development and test sets. This contrasts with the supervised parser, where all the combined training data performs best. Thus, supervised training benefits from broader data, while LLM prompting benefits most from in-domain demonstrations.

Finally, raw-text evaluation reveals differences in tokenization robustness. CamelParser shows only a small drop from gold-tokenized to raw-text parsing, and Gemini is similarly stable. In contrast, GPT, Llama, and Fanar suffer larger drops, partly reflecting weaker tokenization F1. These results highlight the importance of evaluating Arabic dependency parsers in both gold-tokenized and raw-text settings.

Variation Across Sources Table 7 shows substantial source variation under gold-tokenized evaluation: ChatGPT is the easiest source for every system, while Spacetoan is the hardest. Spacetoan

Model	Shots	Data	Dev								Test							
			Gold Tokens			Raw Text					Gold Tokens			Raw Text				
			UAS	LS	LAS	Tok	UAS	LS	LAS	UAS	LS	LAS	Tok	UAS	LS	LAS		
CamelParser2.1	–	B	94.2	96.0	92.4	99.3	93.7	95.6	91.9	95.4	97.1	93.9	99.3	94.8	96.6	93.3		
CamelParser2.1	–	P+C	90.7	93.9	89.4	99.3	90.4	93.5	89.0	91.9	94.8	90.5	99.3	91.4	94.4	90.0		
CamelParser2.1	–	P+C+B	95.5	96.9	94.2	99.3	95.1	96.6	93.7	96.8	97.7	95.5	99.3	96.3	97.3	95.0		
Gemini 3.5 Flash	0	–	81.4	83.3	73.8	92.3	72.4	76.1	66.8	82.4	85.6	76.0	95.4	74.5	78.5	69.7		
Gemini 3.5 Flash	10	B	93.6	96.0	91.7	98.5	93.0	95.1	91.2	94.4	96.9	92.9	98.8	93.6	96.0	92.2		
Gemini 3.5 Flash	10	P+C	87.7	92.6	86.0	98.0	87.1	91.7	85.4	88.8	94.0	87.2	97.5	87.3	92.2	85.8		
Gemini 3.5 Flash	10	P+C+B	93.3	95.9	91.5	98.5	93.0	95.2	91.2	94.0	96.8	92.5	98.5	92.9	95.7	91.5		
GPT-5.4	0	–	70.7	72.1	56.0	93.0	64.8	74.2	55.5	71.3	73.8	57.6	91.6	66.4	74.3	57.1		
GPT-5.4	10	B	91.0	93.5	87.9	98.4	88.8	92.9	86.1	91.8	94.3	88.9	97.9	89.1	92.7	86.3		
GPT-5.4	10	P+C	84.1	89.5	80.6	97.2	82.2	87.9	79.0	84.5	90.8	81.2	95.9	81.9	87.4	78.5		
GPT-5.4	10	P+C+B	90.7	93.5	87.7	98.3	88.2	92.7	85.7	91.6	94.3	88.7	97.8	88.2	92.4	85.5		
Llama-4-Scout	0	–	44.5	51.5	27.0	50.5	36.2	41.0	23.0	45.6	52.1	28.1	51.4	36.1	40.3	23.3		
Llama-4-Scout	10	B	82.3	87.9	76.9	90.5	78.5	83.3	74.1	82.6	88.3	77.2	89.4	77.4	81.5	72.6		
Llama-4-Scout	10	P+C	73.8	82.7	68.0	78.8	68.3	73.7	62.6	72.8	83.0	66.4	78.6	66.2	71.2	60.4		
Llama-4-Scout	10	P+C+B	82.1	87.9	76.9	88.2	77.5	82.7	73.2	82.0	88.3	76.5	88.4	76.4	81.0	71.7		
Fanar-2	0	–	34.9	34.1	14.9	43.8	26.3	31.1	14.2	35.8	34.6	15.9	46.8	26.1	30.5	14.4		
Fanar-2	10	B	78.4	84.7	71.8	89.4	74.8	80.2	68.8	79.2	85.2	72.0	88.8	75.4	80.0	69.3		
Fanar-2	10	P+C	67.6	78.6	60.6	78.9	62.6	71.1	56.1	68.7	79.0	60.7	79.0	62.1	69.8	55.1		
Fanar-2	10	P+C+B	78.0	84.8	71.4	88.8	74.3	80.5	68.8	78.5	84.7	71.3	87.6	74.0	79.4	68.0		

Table 6: **Development and test parsing results.** B, C, and P denote BAREC-TB, CamelTB, and PATB, respectively. For CamelParser2.1, Data indicates the training data. For LLMs, Data indicates the retrieval pool for 10-shot prompting. **Bold** indicates the best result in each metric column; dotted underline indicates the best result within each LLM block. Ties are marked at the displayed precision.

Source	<i>n</i>	CP2.1	Gemini	GPT	Llama	Fanar
Curriculum	804	95.5	92.9	90.1	83.6	76.7
Majed	976	95.1	91.4	85.9	71.7	65.8
Green Library	196	95.9	92.6	88.8	74.2	71.8
Hindawi	297	94.7	93.2	87.7	69.8	65.6
ArabicMMLU	171	96.9	96.2	94.4	88.7	84.5
Other	39	93.4	90.7	86.5	69.3	61.0
Spaceteon	97	86.1	85.3	76.2	62.3	50.5
ChatGPT	152	99.6	99.2	98.2	94.1	94.7

Table 7: **Gold-tokenized test LAS by source.** CP2.1 denotes CamelParser2.1, trained on PATB, CamelTB, and BAREC-TB; the LLM results use ten in-context examples retrieved from BAREC-TB. *n* denotes the number of test sentences.

consists of very short song-lyric fragments (4.3 tokens on average), providing little context for resolving attachments; 54% of its lines contain no verb, removing a central cue for clause structure; and 28% of its surface tokens are unseen in the BAREC-TB training data, compared with 5–16% for the other sources. Together, these properties create substantial syntactic and lexical difficulty for the parsers. Appendix C reports the full gold-tokenized and raw-text source-wise metrics.

7 Conclusion and Future Work

We presented BAREC-TB, the first dependency treebank for Arabic children’s texts. The corpus contains approximately 156K words from curricula, reading materials, stories, comics, poems, and songs, and is annotated following the CATiB dependency representation. Our analyses show that BAREC-TB complements existing Arabic treebanks by covering lower-readability, child-directed text with shorter sentences, simpler tree structures, and a more child-oriented lexical profile. We also established parsing benchmarks with a trainable parser and LLMs, where the best performance is achieved by CamelParser trained on PATB, CamelTB, and BAREC-TB.

In future work, we plan to expand BAREC-TB by annotating more children’s and educational texts from a broader range of genres and reading levels. We also aim to use the treebank for syntax-aware readability assessment and other educational NLP applications, including text simplification and automatic text leveling.

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Limitations

Although BAREC-TB provides a new resource for Arabic children’s dependency parsing, it remains relatively small compared with larger Arabic treebanks. Its source coverage is also limited, and therefore may not fully represent the diversity of Arabic children’s texts across age groups, curricula, and writing styles. In addition, BAREC-TB does not currently include dialectal data, although dialectal language may appear in some children’s texts, especially in dialogue, songs, comics, and regionally oriented materials. We acknowledge that the parsers benchmarked in this work may produce errors, which can propagate to downstream educational NLP applications when parsing is used as a preprocessing step, including syntax-based readability assessment or essay scoring systems.

Ethics Statement

All annotators are paid fair wages for their contribution. The data will be openly available to promote transparency, reproducibility, and collaboration in Arabic language research. We used AI writing assistance within the scope of “Assistance purely with the language of the paper” described in the ACL Policy on Publication Ethics.

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A Licenses

We list below the licenses of the data and tools used in this work, all of which are employed in accordance with their intended use.

- Penn Arabic Treebank (Maamouri et al., 2004) Part 1 v 4.1 (LDC2010T13), Part 2 v 3.1 (LDC2011T09), and Part 3 v 3.2 (LDC2010T08): LDC User Agreement for Non-Members.

- Camel Treebank (Habash et al., 2022): Non Commercial Use - NYUAD License.¹⁰
- SAMER Lexicon (Al Khalil et al., 2020): Non Commercial Use - NYUAD License.¹¹
- BAREC Corpus (Elmadani et al., 2025b): Creative Commons Attribution Share Alike 4.0.
- BAREC-10M Corpus (Elmadani et al., 2026): Creative Commons Attribution Share Alike 4.0.
- CAMEL Tools (Obeid et al., 2020) and CAMELBERT (Inoue et al., 2021): MIT License.
- SAMA3.1 (Graff et al., 2009): LDC Standard Arabic Morphological Analyzer (SAMA) Version 3.1 Agreement.¹²
- AraBERTv2+D3Tok+Reg (Elmadani et al., 2025b): MIT License.
- CamelParser2.1 (Adel et al., 2026): MIT License.
- Fanar-2-27B-Instruct (TEAM et al., 2026): Apache License 2.0.¹³
- Llama-4-Scout-17B-16E-Instruct: Llama 4 Community License Agreement.¹⁴

B Model Sizes, Hyperparameters, and Computational Budget

This appendix reports model sizes, access methods, training hyperparameters, and computational budget for the parsing experiments. Supervised-parser training and decoding used one NVIDIA A100 GPU per run, while Llama-4-Scout and Fanar-2 inference used four NVIDIA A100 GPUs with tensor parallelism.. The supervised CamelParser model consists of a CamelBERT-MSA encoder and a SuPar biaffine dependency parser head, totaling approximately 110.3M parameters. Llama-4-Scout is a mixture-of-experts model with 109B total parameters and 17B active parameters, and was served using vLLM. Fanar-2 has 27B parameters and was also served using vLLM. GPT-5.4 was accessed through the OpenAI Batch API, and Gemini 3.5 Flash was accessed through the Gemini API. Since GPT-5.4 and Gemini 3.5 Flash are closed API models, their parameter counts are not publicly disclosed. We report API costs for proprietary models

¹⁰<http://treebank.camel-lab.com/>

¹¹<https://camel.abudhabi.nyu.edu/samer-readability-lexicon/>

¹²<https://catalog.ldc.upenn.edu/LDC2010L01>

¹³<https://huggingface.co/QCRI/Fanar-2-27B-Instruct>

¹⁴<https://huggingface.co/meta-llama/Llama-4-Scout-17B-16E-Instruct/blob/main/LICENSE>

Parameter	Value	Parameter	Value
bert_pooling	mean	mlp_dropout	0.33
clip	5.0	mu	0.9
encoder_dropout	0.1	n_arc_mlp	500
epochs	10	n_bert_layers	4
eps	1e-08	n_rel_mlp	100
fix_len	20	nu	0.999
lr	5e-5	update_steps	4
lr_rate	10	warmup	0.1
min_freq	2	weight_decay	0
mix_dropout	0.0		

Table 8: Hyperparameters used to train all supervised parsing models.

Model	Data	Train Time min.
CamelParser2.1	B	2.7
CamelParser2.1	P+C	10.7
CamelParser2.1	P+C+B	12.4
Total supervised training		25.9

Table 9: **Training time for supervised parsing models.** B, C, and P denote BAREC-TB, CamelTB, and PATB, respectively. Training time is reported in minutes.

and runtime only for locally served models; we do not estimate institutional GPU electricity or amortized hardware cost.

Table 8 lists the hyperparameters used for all supervised CamelParser training runs. The same configuration is used across the three training-data settings.

Table 9 reports supervised parser training time. Training the three CamelParser variants required 25.9 minutes in total, corresponding to approximately 0.4 GPU hours on a single NVIDIA A100.

Table 10 reports evaluation runtime and API cost for all benchmark runs, broken down by split and input condition. Across all evaluation runs, the benchmark required 8,178.2 minutes, or approximately 136.3 hours, with a total API cost of \$1,085.03. The monetary cost comes from proprietary API-based models; locally served models are reported with zero API cost.

C Full Source-wise Parsing Results

Table 11 provides the full metrics underlying the gold-tokenized LAS comparison in Table 7; Table 12 reports the corresponding raw-text results.

Model	Shots	Data	Dev				Test			
			Gold Tokens		Raw Text		Gold Tokens		Raw Text	
			Time min.	Cost USD	Time min.	Cost USD	Time min.	Cost USD	Time min.	Cost USD
CamelParser2.1	–	B	0.1	0.00	2.5	0.00	0.1	0.00	2.6	0.00
CamelParser2.1	–	P+C	0.1	0.00	2.5	0.00	0.1	0.00	2.6	0.00
CamelParser2.1	–	P+C+B	0.1	0.00	2.5	0.00	0.1	0.00	2.6	0.00
Gemini 3.5 Flash	0	–	16.8	31.86	51.2	46.11	26.8	31.65	52.6	45.87
Gemini 3.5 Flash	10	B	25.0	40.00	50.8	48.74	30.1	39.73	30.5	48.33
Gemini 3.5 Flash	10	P+C	27.9	44.95	29.8	54.48	35.1	44.33	19.8	53.47
Gemini 3.5 Flash	10	P+C+B	50.3	41.08	45.2	48.94	50.1	40.21	18.1	48.73
GPT-5.4	0	–	600.9	7.67	207.8	8.88	602.9	7.69	207.7	8.73
GPT-5.4	10	B	578.6	25.81	201.8	27.17	210.0	26.44	522.1	27.50
GPT-5.4	10	P+C	589.2	31.54	573.6	32.17	605.2	31.44	204.9	31.83
GPT-5.4	10	P+C+B	202.7	26.55	219.2	27.73	586.4	26.99	213.1	28.39
Llama-4-Scout	0	–	29.1	0.00	33.1	0.00	29.4	0.00	31.8	0.00
Llama-4-Scout	10	B	41.7	0.00	45.4	0.00	45.5	0.00	49.1	0.00
Llama-4-Scout	10	P+C	54.2	0.00	57.1	0.00	58.9	0.00	60.7	0.00
Llama-4-Scout	10	P+C+B	44.9	0.00	48.8	0.00	50.7	0.00	53.4	0.00
Fanar-2	0	–	19.4	0.00	23.4	0.00	20.1	0.00	22.6	0.00
Fanar-2	10	B	29.4	0.00	35.7	0.00	32.1	0.00	41.4	0.00
Fanar-2	10	P+C	37.1	0.00	43.4	0.00	40.2	0.00	48.4	0.00
Fanar-2	10	P+C+B	31.3	0.00	39.5	0.00	35.0	0.00	43.0	0.00
All runs (sum)			2,378.8	249.47	1,713.2	294.21	2,459.0	248.50	1,627.2	292.85

Table 10: **Evaluation runtime and cost.** B, C, and P denote BAREC-TB, CamelTB, and PATB, respectively. Time is reported in minutes and cost in USD for each split and input condition. API-model costs reflect the charges at the time of the experiments; locally run models report 0.00 USD because no API charge was incurred. CamelParser2.1 used one NVIDIA A100 GPU, while Llama and Fanar used four. The final row sums all runs in each column.

Source	CamelParser2.1			Gemini			GPT			Llama			Fanar		
	UAS	LS	LAS	UAS	LS	LAS	UAS	LS	LAS	UAS	LS	LAS	UAS	LS	LAS
Curriculum	96.4	97.7	95.5	93.9	97.1	92.9	92.5	95.1	90.1	87.6	91.6	83.6	82.5	87.8	76.7
Majed	96.4	97.6	95.1	93.2	96.0	91.4	89.7	92.6	85.9	78.6	85.1	71.7	74.6	81.5	65.8
Green Library	96.9	98.5	95.9	93.9	97.5	92.6	91.1	95.6	88.8	78.9	89.8	74.2	78.3	87.5	71.8
Hindawi	96.0	97.4	94.7	94.7	97.5	93.2	90.9	94.2	87.7	76.1	84.9	69.8	73.4	82.8	65.6
ArabicMMLU	98.2	98.2	96.9	97.5	97.8	96.2	97.0	96.4	94.4	92.9	93.3	88.7	90.5	90.7	84.5
Other	98.6	94.1	93.4	95.9	93.5	90.7	90.3	92.8	86.5	76.8	84.9	69.3	69.5	81.5	61.0
Spacetoan	92.3	88.2	86.1	87.3	90.6	85.3	80.3	82.2	76.2	68.8	74.0	62.3	58.9	63.9	50.5
ChatGPT	100.0	99.6	99.6	99.8	99.4	99.2	98.4	99.0	98.2	94.4	95.4	94.1	95.8	96.5	94.7

Table 11: **Full gold-tokenized test results by source.** CamelParser2.1 is trained on PATB, CamelTB, and BAREC-TB; the LLM results use ten in-context examples retrieved from BAREC-TB. Bold indicates the best result in each row for each metric.

Source	CamelParser2.1				Gemini				GPT				Llama				Fanar			
	Tok	UAS	LS	LAS	Tok	UAS	LS	LAS	Tok	UAS	LS	LAS	Tok	UAS	LS	LAS	Tok	UAS	LS	LAS
Curriculum	99.2	95.8	97.2	94.8	99.5	94.1	97.0	93.1	99.1	91.7	95.3	89.9	92.9	85.0	89.3	81.9	91.8	82.5	86.4	78.1
Majed	99.4	96.1	97.3	94.9	99.1	92.7	95.5	90.9	98.1	87.3	91.3	83.6	88.6	74.1	78.1	67.8	87.7	70.0	74.8	61.9
Green Library	99.2	96.4	98.1	95.4	99.1	93.4	97.0	92.1	98.5	89.1	94.4	86.7	85.4	71.9	79.2	67.3	86.0	70.6	80.3	65.3
Hindawi	98.9	95.1	96.6	93.7	95.6	90.9	93.9	89.9	94.5	84.0	89.7	81.2	84.5	68.7	76.0	63.7	83.8	67.7	75.0	61.2
ArabicMMLU	99.7	97.8	97.9	96.6	99.3	96.8	97.2	95.5	98.2	93.5	93.9	90.9	91.7	84.4	84.1	80.5	92.5	87.0	86.5	82.2
Other	100.0	98.4	94.1	93.2	99.8	95.5	93.2	90.1	98.9	89.4	91.9	84.9	90.0	73.1	79.7	65.7	83.2	71.6	76.8	63.4
Spacetoan	97.5	91.3	86.5	84.6	99.3	85.3	88.2	83.7	97.0	75.7	80.0	71.4	93.0	73.6	76.0	67.5	88.1	66.3	68.3	58.2
ChatGPT	100.0	100.0	99.6	99.6	100.0	99.9	99.9	99.8	100.0	99.0	99.3	98.7	98.7	93.6	94.3	92.6	97.4	92.3	93.2	90.8

Table 12: **Full raw-text test results by source.** CamelParser2.1 is trained on PATB, CamelTB, and BAREC-TB; the LLM results use ten in-context examples retrieved from BAREC-TB. Bold indicates the best result in each row for each metric.